Exercise J01

Q1. f is a continuous function on the real line. Given that
$x^2 + (f(x) - 2)\ \ x - \sqrt{3}\ .\ f(x) + 2\sqrt{3} - 3 = 0$. Then the value of $f(\sqrt{3})$

(A) can not be determined (B) is $2\ (1 - \sqrt{3})$ (C) is zero (D) is $\frac{2(\sqrt{3}-2)}{\sqrt{3}}$

Ans. B

Sol. $(x-\sqrt{3})f(x) = -x^2 + 2x - 2\sqrt{3} + 3$

$$f(x) = \frac{-x^2 + 2x - \sqrt{3} + 3}{x - \sqrt{3}}$$

$$= \frac{(x-\sqrt{3})(2-\sqrt{3}-x)}{x-\sqrt{3}} = 2 - \sqrt{3} - x$$

$f(\sqrt{3}) = 2 - 2\sqrt{3}$

Q2. The function $f(x) = [x]^2 - [x^2]$ (where [y] is the greatest integer less than or equal to y), is discontinuous at :

(A) all integers (B) all integers except 0 & 1 (C) all integers except 0

(D) all integers except 1

Ans. D

Sol. $f(x) = [x]^2 - [x^2]$

$(-1)^2 - 0 = 1$ $\qquad x = 0 - h$

$$f(x) = \begin{cases} 0 & x = 0 \\ 0 - 0 = 0 & 0 < x < 1 \\ 1^2 - 1 = 0 & 1 \le x < \sqrt{2} \\ 1 - 2 = -1 & \sqrt{2} \le x < \sqrt{3} \\ 1 - 3 = -2 & \sqrt{3} \le x < 2 \end{cases}$$

$[x] = -1$

$[x^2] = 0$

$1 \le x^2 < 2$

$[x] = 1$

$2 \le x^2 < 3$

i.e. it is continuous at x = 1 only and discontinuous at all other points]

Q3. Let g(x) = $\tan^{-1}|x| - \cot^{-1}|x|$, $f(x) = \frac{[x]}{[x+1]}\{x\}$, h(x) = |g(f(x))| where {x} denotes fractional part and [x] denotes the integral part then which of the following holds good?

(A) h is continuous at x = 0 (B) h is discontinuous at x = 0 (C) $h(0^-) = \pi/2$

(D) $h(0^+) = -\pi/2$

Ans. A

Sol. domain of f(x), $x \in R - [-1, 0)$

Continuity at x = 0

h(0) = |g(f(0))|

= |g(0)| f(0) = 0

$= |0 - \frac{\pi}{2}|$

$h(0) = \frac{\pi}{2}$

h(0 + h) = |g(f(0 + h))| $\mathrm{f(0+h)} = \frac{[0+\mathrm{h}]}{[0+\mathrm{h}+1]} = 0$

= |g(0)|

$= \frac{\pi}{2}$

h(0 –h) = |g(f(0 – h))|

= need

not to calculate (0 – h) is not in domain of f(x)

hence R.H.L. = g(0)

so function is continuous at $x = \frac{\pi}{2}$

Q4. The function f(x) = [x].$cos\frac{2x-1}{2}\pi$, where [•] denotes the greatest integer function, is discontinuous at

(A) all x (B) all integer points (C) no x (D) x which is not an integer

Ans. C

Sol. $f(x) = [x]cos\left(\frac{2x-1}{2}\right)\pi$

Let check at integer x = 1

$f(1) = [1]cos\left(\frac{1}{2}\right)\pi = 0$

$f(1+h) = [1+h]cos\left(\frac{2+2h-1}{2}\right)\pi = 0$

$$f(1-h) = [1-h]cos\left(\frac{2-2h-1}{2}\right)\pi = 0$$

Q5. Consider the functions

f (x) = sgn (x – 1) and g (x) = cot–1[x – 1]

where [ ] denotes the greatest integer function.

Statement-1 : The function F (x) = f (x) · g (x) is discontinuous at x = 1.

because

Statement-2 : If f (x) is discontinuous at x = a and g (x) is also discontinuous at x = a then the product function f (x) · g (x) is discontinuous at x = a.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Ans. 5f15366eb58d37a192825007

Sol. F(x) = f(x).g(x)

$F(x) = sgn(x–1).\ cot^{-1}[x–1]$

continuity at x = 1

F(1) = 0

$F(1 + h) = sgn(1 + h – 1).cot^{-1}[1+ h–1]$

$= sgn(h).cot^{-1}[h]$

= 0

$F(1 – h) = sgn(1 – h – 1).cot^{-1}[1 – h – 1]$

$= sgn(–h).cot^{-1}[–h\ ]$

$= -1 \times \cot^{-1} = \frac{\pi}{4}$

hence F(x) is discontinuous at x = 1.

Statement (2) is wrong as we have studied in concepts.

Q6. $$f(x) = \begin{bmatrix} \sin\left(\frac{a-x}{2}\right)\tan\left[\frac{\pi x}{2a}\right] & for\ x > a \\ \frac{\left[\cos\left(\frac{\pi x}{2a}\right)\right]}{a-x} & for\ x < a \end{bmatrix}$$

where [x] is the greatest integer function of x, and a > 0, then

(A) $f(a^{-}) < 0$ (B) f has a removable discontinuity at x = a

(C) f has an irremovable discontinuity at $x = a$ (D) $f(a^+) < 0$

Ans. B

Sol. $\mathrm{R.H.L.} = \sin\left(\frac{a-a-h}{2}\right)\tan\left[\frac{\pi}{2a}(a+h)\right]$

$= \sin\left(\frac{-h}{2}\right)\tan\left[\frac{\pi}{2}+\frac{\pi h}{2a}\right]$

$= 0$

$\mathrm{L.H.L.} = \frac{\left[\cos\frac{\pi}{2a}(a-h)\right]}{a-(a-h)}$

$= \frac{\left[\cos\left(\frac{\pi}{2}-\frac{\pi h}{2a}\right)\right]}{h}$

$= \frac{\left[\sin\frac{\pi h}{2a}\right]}{h}$

L.H.L. = 0

hence f(x) is removable discontinuity at $x = a$.

Q7. Let $f : R \to R$ be a function defined by $f(x) = \text{Min}\ \{x + 1, |x| + 1\}$. Then which of the following is true ?

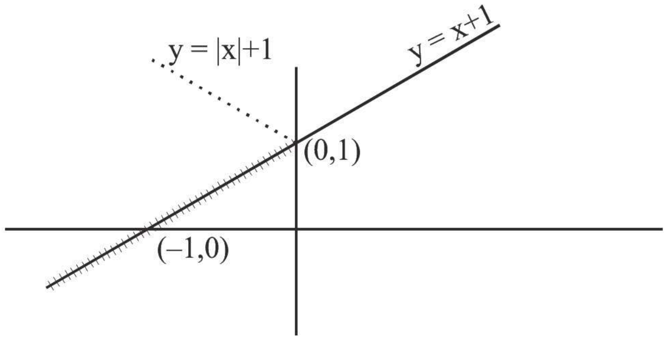
(A) $f(x) \geq$ for all $x \in R$ (B) f(x) is not differentiable at $x = 1$

(C) f(x) is differentiable everywhere (D) f(x) is not differentiable at $x = 0$

Ans. C

Sol. $f(x) = \min\{x + 1, |x| + 1\}$

$y = x + 1 \qquad y = |x| + 1$



$$f(x) = \min\{x + 1, |x| + 1\} = \begin{cases} 1+x & ; -\infty < x \leq 0 \\ x+1 & ; x > 1 \end{cases}$$

hence

$f(x) = 1 + x$; x R

so f(x) is differentiable everywhere.

Q8. Let $f(x) = \begin{cases} (x-1)\sin\dfrac{1}{x-1} & \text{if } x \neq 1 \\ 0 & \text{if } x = 1 \end{cases}$. Then which one of the following is true ?

(A) f is differentiable at x = 0 and at x = 1

(B) f is differentiable at x = 0 but not at x = 1

(C) f is differentiable at x = 1 but not at x = 0

(D) f is neither differentiable at x = 0 nor at x = 1

Ans. B

Sol. $$\text{R.H.D} = \lim_{h\to\infty} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h\to 0} \frac{h\sin\left(\frac{1}{h}\right) - 0}{h}$$

$$= \lim_{h\to 0} \sin\left(\frac{1}{h}\right) = \text{DNE}$$

hence not differentiable at x = 1

differentiability at x = 0

$$\text{R.H.D} = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h\to 0} \frac{(h-1)\sin\left(\frac{1}{h-1}\right) - (-1)\sin\left(\frac{1}{-1}\right)}{h}$$

$$= \lim_{h\to 0} \frac{(h-1)\sin\left(\frac{1}{h-1}\right) - \sin 1}{h} \qquad \left(\frac{0}{0}\right)$$

using L'hospital

$$= \frac{(1)\sin\left(\frac{1}{h-1}\right) + (h-1)\cos\frac{1}{(h-1)}\left(-\frac{1}{(h-1)^2}\right) - 0}{1}$$

= sin1 – cos1

$$\text{L.H.D} = \lim_{h\to 0} \frac{(0-h-1)\sin\left(\frac{1}{0-h-1}\right) - 0}{h}$$

= $\left(\frac{0}{0}\right)$ hecne using L'hospital

= sin1 – cos1

hence function is derivable at x = 0

Q9. Let $g(x) = \dfrac{(x-1)^n}{ln \cos^m(x-1)}$; $0 < x < 2$, m and n are integers, $m \neq 0$, $n > 0$ and let p be the left hand derivative of | x – 1 | at x = 1. If $\lim\limits_{x\to 1^+} g(x) = p$, then value of n + m is :

Ans. 1

Sol. P = 2

put x = 1 + h, $h \to 0$

$$\lim_{h\to 0} \frac{h^n}{\text{m ln cosh}} = 1$$

Diff. w.r.t. h

$$-\frac{n}{m}\frac{h^{n-1}}{\tan h} = 1$$

n = 1

$$\lim_{h\to 0} \frac{-2}{m\left(\frac{\tan h}{h}\right)} = 2$$

m = –1

n + m = 1

Q10. For what triplets of real numbers (a, b, c) with $a \neq 0$ the function

$f(x) = \begin{cases} x & x \leq 1 \\ ax^2 + bx + c & \text{otherwise} \end{cases}$ is differentiable for all real x ?

(A) $\{(a, 1-2a, a) \mid a \in R, a \neq 0\}$ (B) $\{(a, 1-2a, c) \mid a, c \in R, a \neq 0\}$

(C) $\{(a, b, c) \mid a, b, c \in R, a + b + c = 1\}$ (D) $\{(a, 1-2a, 0) \mid a \in R, a \neq 0\}$

Ans. A

Sol.
$$f(x) = \begin{cases} x & , \ x \leq 1 \\ ax^2+bx+c & , \ \text{otherwise} \end{cases}$$

f(x) should be continous at x =1

it gives $a + b + c = 1$

f(x) should be differentiable at x = 1

it gives $2a + b = 1 \Rightarrow b = 1 - 2a \quad c = 1 - a - b = a$

Q11. Number of points of non-differentiability of the function $g(x) = [x^2]\{\cos^2 4x\} + \{x^2\}[\cos^2 4x] + x^2 \sin 2\ 4x + [x^2]\ [\cos^2 4x] + \{x^2\}\{\cos^2 4x\}$ in $(-50, 50)$ where [x] and {x} denote the greatest integer function and fractional part function of x respectively,
is equal to

(A) 98 (B) 99 (C) 100 (D) 0

Ans. D

Sol. $g(x) = ([x^2] + \{x^2\}) + x^2 \sin^2 4x = x^2 \cos 24x + x^2 \sin^2 4x = x^2$

Clearly g(x) is continuous and differenitable $\forall\ x \in R$.]

Q12. If $f(x) = \begin{bmatrix} x + \{x\} + x\sin\{x\} & for x \neq 0 \\ 0 & for x = 0 \end{bmatrix}$ where {x} denotes the fractional part function, then :

(A) 'f' is continuous & diff. at x = 0 (B) 'f' is continuous but not diff. at x = 0

(C) 'f' is continuous & diff. at x = 2 (D) none of these

Ans. D

Sol. $f(0) = \lim_{x\to\infty} f(x) = 0 - 1 + 0.\sin(-1) = -1$

$f(0^+) = \lim_{x\to\infty} f(x) = 0 + 0 + 0.\sin 0 = f(0)$

f(x) is not continuous at x = 0 at x = 2,

$f(2+) = 2 + 2 + 2 \sin 2 = 4 + 2 \sin 2$

$f(2-) = 2 + 1 + 2 \sin 1 = 3 + 2 \sin 1$

f(x) is not continuous at x = 2

Q13. The set of all points where the function $f(x) = \frac{x}{1+|x|}$ is differentiable is :

(A) $(-\infty, \infty)$ (B) $[0, \infty)$ (C) $(-\infty, 0) \cup (0, \infty)$ (D) $(0, \infty)$

Ans. A

Sol. $f(x) = \begin{cases} \frac{x}{1-x} & , x < 0 \\ \frac{x}{1+x} & , x \geq 0 \end{cases}$

f(x) is continous and differntiable for $x \in R$

$f(x) = \begin{cases} |x| & , |x| < 1 \\ 0 & , |x| = 1 \\ |x| & , |x| > 1 \end{cases}$

f(x) is discontinous at $|x| = 1$

Q14. Consider the function $f(x) = \begin{bmatrix} x\{x\}+1 & 0 \leq x < 1 \\ 2-\{x\} & 1 \leq x \leq 2 \end{bmatrix}$ where {x} denotes the fractional part function. Which one of the following statements is NOT correct?

(A) $\lim_{x\to 1} f(x)$ exists (B) $f(0) \neq f(2)$ (C) f (x) is continuous in [0, 2] (D) None

Ans. C

Sol. $f(1^+) = f(1^-) = f(1) = 2$

$f(0) = 1, \quad f(2) = 2$

$f(2^-) = 1; \quad f(2) = 2 \quad \Rightarrow \quad$ f is not continuous at x = 2 Ans.]

Q15. Let $g : R \to R$ be a differentiable function such that $g(2) = -40$ and $g'(2) = -5$. Then $\lim_{x\to 0}\left(\frac{g(2-x^2)}{g(2)}\right)^{\frac{4}{x^2}}$ is equal to

(A) e^{32} (B) $\sqrt{e}$ (C) $\frac{1}{\sqrt{e}}$ (D) e^{-5}

Ans. C

Sol. $\lim_{x\to 0}\left(\frac{g(2-x^2)}{g(2)}\right)^{\frac{4}{x^2}}$

$(1^{\infty} form) = \lim_{x\to 0} 4\left(\frac{g(2-x^2)-g(2)}{x^2}\right) \times \frac{1}{g(2)} = e^{-4\frac{g(2)}{g(2)}} = e^{\frac{-4x-5}{-40}} = e^{-\frac{1}{2}} = \frac{1}{\sqrt{e}}$

Q16. Let $f : (0, 1) \to R$ be defined by $f(x) = \frac{b-x}{1-bx}$ where b is a constant such that 0 < b < 1. Then –

(A) f is not invertible on (0, 1) (B) $f \neq f^{-1}$ on (0, 1) and $f'(b) = \frac{1}{f'(0)}$

(C) $f = f^{-1}$ on (0, 1) and $f'(b) = \frac{1}{f'(0)}$ (D) f^{-1} is differentiable on (0, 1)

Ans. A

Sol. $f : (0,1) \to R$

$f(x) = \dfrac{b - x}{1 - bx}$ $\qquad b \in (0, 1)$

$\Rightarrow f'(x) = \dfrac{b^2 - 1}{(bx - 1)^2}$

$\Rightarrow f'(x) < 0 \forall x \in (0, 1)$

hence f(x) is decreasing function

hence its range (–1, b)

$\Rightarrow$ co-domain $\neq$ range

$\Rightarrow$ f(x) is non-invertible function

Q17. Let S = {t $\in$ R: f(x) = |x – π| . ($e^{|x|}$ – 1) sin |x| is not differentiable at t}. Then the set S is equal to

(A) $\{\pi\}$ (B) $\{0, \pi\}$ (C) ϕ (an empty set) (D) $\{0\}$

Ans. C

Sol.

$$f(x) \begin{cases} \rightarrow (x - \pi)(e^x - 1)\sin x & x \geq \pi \\ \rightarrow (\pi - x)(e^x - 1)\sin x & 0 \leq x < \pi \\ \rightarrow -(\pi - x)(e^x - 1)\sin x & x < 0 \end{cases}$$

at x = 0

$f'(0^+) = 0$

$f'(0^-) = 0$

at x $= \pi$

$f'(\pi^+) = 0$

$f'(\pi^-) = 0$

It is differentiable at x = 0, ϕ

S = { } = ϕ

Q18. Let f be an injective and differentiable function such that f (x) $\cdot$ f (y) + 2 = f (x) + f (y) + f (xy) for all non negative real x and y with f '(0) = 0, f '(1) = 2 $\neq$f (0), then

(A) x f '(x) – 2 f (x) + 2 = 0 (B) x f '(x) + 2 f (x) – 2 = 0 (C) x f '(x) – f (x) + 1 = 0

(D) 2 f (x) = f '(x) + 2

Ans. A

Sol. f(x). f(y) + 2 = f(x) + f(y) + f(xy)

differentiaable w.r.t. ‘y’

f(x) f' (x) = f '(1) = (1) + x f ' (x)

2f '(x) = 2 + x f ' (x)

x f ' (x) – 2f(x) + 2 = 0

Q19. Let f (x) be continuous and differentiable function for all reals.
f (x + y) = f (x) – 3xy + f (y). If $\lim_{h\to 0} \frac{f(h)}{h} = 7$, then the value of f ' (x) is

(A) -3x (B) 7 (C) –3x +7 (D) 2 f (x) + 7

Ans. C

Sol. $f'(x) = \frac{f(x+h)-f(x)}{h} = \frac{f(h)-3xh}{h} = 7 - 3x$

Q20. If f(x + y) = f(x) + f(y) + c, for all real x and y and f(x) is continuous at x = 0 and f ' (0) = 1 then f ' (x) equals to

(A) c (B) –1 (C) 0 (D) 1

Ans. D

Sol. $f'(0) = \lim_{h\to 0} \frac{f(0+h)-f(0)}{h} = 1$ also f(0) = – c

$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{f(x)+(h)+c-f(x)}{h}$

$= \lim_{h\to 0} \frac{f(h)-f(0)}{h} = f'(0) = 1$

$\therefore$ f ' (x) = 1

Q21. If for all values of x & y; $f(x + y) = f(x) .f(y)$ and $f(5) = 2$, $f'(0) = 3$, then $f'(5)$ is-

(A) 3 (B) 4 (C) 5 (D) 6

Ans. D

Sol. $f(x + y) = f(x) . f(y) \quad \forall x, y$

$\therefore f(5 + 0) = f(5) . f(0) \quad \{\because f(5) = 2$

$\therefore f(0) = 1$

Now $f'(5) = \lim\limits_{h\to 0} \dfrac{f(5+h) - f(5)}{h}$

$= \lim\limits_{h\to 0} \dfrac{f(5)f(h) - f(5)}{h}$

$= f(5) \lim\limits_{h\to 0} \dfrac{f(h) - f(0)}{h}$

$= f(5)\, f'(0) = 2 \times 3 \Rightarrow 6$

Q22. If f is a real-valued differentiable function satisfying $| f(x) – f (y) | \le (x – y)^2$, $x, y \in R$ and $f(0) = 0$, then $f(1)$ equals

(A) –1 (B) 0 (C) 2 (D) 1

Ans. B

Sol. $|f(x) – f(y)| \le |x – y|^2$

$\Rightarrow \dfrac{|f(x) - f(y)|}{|x - y|} \le |x - y|$

$\Rightarrow \dfrac{|f(x) - f(y)|}{|x - y|} \le \lim\limits_{x\to y} |x - y|$

$\Rightarrow f'(x) \le 0 \Rightarrow f'(x) = 0$

$\Rightarrow$ f(x) is continuous function

$\therefore f(1) = 0 = f(0)$

Q23. Suppose f(x) is differentiable at x = 1 and $\lim\limits_{h\to 0} \dfrac{1}{h} f(1 + h) = 5$, then $f'(1)$ equals -

(A) 3 (B) 4 (C) 5 (D) 6

Ans. C

Sol. Apply L Hospital rule

$\lim\limits_{h\to 0} \dfrac{f'(1+h)}{1} = 5$

$\Rightarrow f'(1) = 5$

Exercise J02

Q1. If the function $f(x) = \dfrac{3x^2 + ax + a + 3}{x^2 + x - 2}$ is continuous at x = – 2. Find f (–2).

Ans. – 1

Sol. $f(x) = \dfrac{3x^2 + ax + a + 3}{(x+2)(x-1)}$

at x = – 2 denominator is zero hence numerator should also be zero.

hence

3(–2)2 + a(–2) + a + 3 = 0

12 – 2a +a +3 = 0

a = 15

$f(x) = \dfrac{3x^2 + 15x + 18}{(x+2)(x-1)}$

here

$= f(-2) = \lim\limits_{x \to -2} \dfrac{3(x^2 + 8x + 6)}{(x+2)(x-1)}$

$f(-2) = \dfrac{3(1)}{(-3)}$

$f(-2) = -1$

Q2. Find all possible values of a and b so that f (x) is continuous for all x ∈ R if

$$f(x) = \begin{cases} |ax+3| & if \quad x \le -1 \\ |3x+a| & if \quad -1 < x \le 0 \\ \frac{b\sin 2x}{x} - 2b & if \quad 0 < x < \pi \\ \cos^2 x - 3 & if \quad x \ge \pi \end{cases}$$

Ans. a = 0, b = 1

Sol. at x = –1

f(–1) = R.H.L.

|–a +3| = |–3 + a|

|a –3| = |a –3|

a ∈ R ....(1)

at x = 0

f(0) = R.H.L

$|a| = \lim\limits_{h \to 0} \dfrac{b\sin 2h}{h} - 2b$

$|a| = \lim\limits_{h \to 0} 2b\left(\frac{\sin 2h}{2h}\right) - 2b$

$|a| = 2b - 2b$

$|a| = 0$

$a = 0$ ....(2)

at x = π

L.H.L. = f(π)

$\lim\limits_{h \to 0} \frac{b\sin 2(\pi - h)}{(\pi - h)} - 2b = (-1)^2 - 3$

$\lim\limits_{h \to 0} \frac{b\sin 2h}{\pi - h} - 2b = -2$

b = 1 ....(3)

by (1), (2) and (3) overall solution

a = 0

b = 1

Q3.

The function $f(x) = \begin{cases} \left(\frac{6}{5}\right)^{\frac{\tan 6x}{\tan 5x}} & if \quad 0 < x < \frac{\pi}{2} \\ b + 2 & if \quad x = \frac{\pi}{2} \\ (1 + |\cos x|)^{\left(\frac{a|\tan x|}{b}\right)} & if \quad \frac{\pi}{2} < x < \pi \end{cases}$

Determine the values of 'a' & 'b' , if f is continuous at x = $\pi/2$.

Ans. a = 0 ; b = –1

Sol.

$L.H.L = \lim\limits_{h \to 0} \left(\frac{6}{5}\right)^{\frac{\tan 6\left(\frac{\pi}{2} - h\right)}{\tan 5\left(\frac{\pi}{2} - h\right)}}$

$= \lim\limits_{h \to 0} \left(\frac{6}{5}\right)^{\frac{-\tan 6h}{\cot h}}$

$= \lim\limits_{h \to 0} \left(\frac{6}{5}\right)^{-\tan 6h . \tan h}$

= 1

$f\left(\frac{\pi}{2}\right) = b + 2$

$= L.H.L. = f\left(\frac{\pi}{2}\right)$

1 = b + 2, b = 1

R.H.L.

$= 1 + \left|\cos\left(\frac{\pi}{2} + h\right)\right|^{\left(\frac{a}{b}\left|\tan\left(\frac{\pi}{2} + h\right)\right|\right)}$

$= (1 + |sinh|)^{\frac{a}{b}|coth|}$

$= (1 + (sinh)^{\frac{a}{b}coth}$ (1^{∞})

$= e^{\frac{a}{b}coth(1+sinh-1)}$

$= e^{\frac{a}{b}cosh}$

$= e^{\frac{a}{b}}$

= but b = 1

$= e^{a}$

R.H.L. = L.H.L hence

$e^{a} = 1$ a = 0

a = b = 1

Q4. Suppose that $f(x) = x^3 - 3x^2 - 4x + 12$ and $h(x) = \begin{bmatrix} \frac{f(x)}{x-3} & , & x \neq 3 \\ K & , & x = 3 \end{bmatrix}$ then

(a) find all zeros of f (x)

(b) find the value of K that makes h continuous at x = 3

(c) using the value of K found in (b), determine whether h is an even function.

Ans. (a) –2,2, 3 (b) K = 5 (c) even

Sol. f (x) = (x + 2) (x – 2) (x – 3)

$h(x) = \begin{cases} (x+2)(x-2) & , x \neq 3 \\ k & , x = 3 \end{cases}$ for continuity $k = \lim_{x \to 3} h(x) = 5$

$h(x) = (x + 2)(x - 2) = x^2 - 4$ which is even $\forall x \in R$

Q5. Let $f(x) = \begin{bmatrix} \frac{1-\sin \pi x}{1+\cos 2\pi x}, & x < \frac{1}{2} \\ p, & x = \frac{1}{2} \\ \frac{\sqrt{2x-1}}{\sqrt{4+\sqrt{2x-1}}-2}, & x > \frac{1}{2} \end{bmatrix}$. Determine the value of p, if possible, so that the function is continuous at x=1/2.

Ans. P not possible.

Sol. at $x = \frac{1}{2}$

$$R.H.L = \frac{\sqrt{2\left(\frac{1}{2}+h\right)-1}}{\sqrt{4+\sqrt{2\left(\frac{1}{2}+h\right)-1-2}}}$$

$$= \frac{\sqrt{2h}}{\sqrt{4+\sqrt{4}-2}}$$

$$= \frac{\sqrt{2h}}{\sqrt{4+\sqrt{4}-2}} \times \frac{4+\sqrt{h}+2}{4+\sqrt{h}+2}$$

$$= \frac{\sqrt{2}\sqrt{h}(6+\sqrt{h})}{\sqrt{h}}$$

$$= 6\sqrt{2}$$

$$L.H.L = \frac{1 - sin\left(\pi\left(\frac{1}{2} - h\right)\right)}{1 + cos2\pi\left(\frac{1}{2} - h\right)}$$

$$= \frac{1 - cos\pi h}{1 - cos2\pi h}$$

$$= \frac{2sin^2\left(\frac{\pi h}{2}\right)}{2sin^2(\pi h)}$$

$$= \frac{1}{4}$$

hence noe possible value of 'p' exists.

Q6. Given the function g (x) = $\sqrt{6-2x}$ and h (x) = $2x^2 - 3x + a$. Then

(a) evaluate h (g(2)) (b) If f (x) = $\begin{bmatrix} g(x), & x \le 1 \\ h(x), & x > 1 \end{bmatrix}$, find 'a' so that f is continuous.

Ans. (a) $4 - 3\sqrt{2} + a$, (b) a = 3

Sol. (a) h(g(2))

$= h(\sqrt{2})$

$= 2(\sqrt{2})^2 - 3(\sqrt{2}) + a$

$= 4 - 3\sqrt{2} + a$

(b) for f(x) at x = 1

f(1) = R.H.L.

g(1) = h(1 + h)

2 = 2 –3 + a

a = 1

Q7. Determine the values of a, b & c for which the function f (x) =

$$\left[\begin{array}{ll} \frac{\sin(a+1)x + \sin x}{x} & for \quad x < 0 \\ c & for \quad x = 0 \\ \frac{\left(x+bx^2\right)^{1/2} - x^{1/2}}{bx^{3/2}} & for \quad x > 0 \end{array}\right.$$

is continuous at x = 0.

Ans. a = – 3/2, b $\neq$ 0, c = 1/2

Sol. $\lim_{x\to 0^-} \frac{\sin(a+1)x + \sin x}{x} = a + 2$

and $\lim_{x\to 0^-} \frac{x + bx^2 - x}{bx^{\frac{3}{2}}\left(\sqrt{x+bx^2}+\sqrt{x}\right)} = \frac{1}{2}$ as $b \neq 0$

according to question

$c = \frac{1}{2} \& a + 2 = \frac{1}{2} \Rightarrow a = \frac{-3}{2}$

Q8. If $f(x) = \frac{\sin 3x + A\sin 2x + B\sin x}{x^5}$ ($x \neq 0$) is cont. at x = 0. Find A & B. Also find f(0).

Ans. A = –4 , B = 5, f(0) = 1

Sol. $f(0) = \lim_{x\to 0} \frac{\sin 3x + A\sin 2x + B\sin x}{x^5}$

$$f(0) = \lim_{x\to 0} \frac{\left(3x - \frac{(3x)^3}{!3} + \frac{(3x)^5}{!5}\ldots\ldots\right) + A\left(2x - \frac{(3x)^3}{!3} + \frac{(3x)^5}{!5}\ldots\ldots\right) + B\left(x - \frac{x^3}{!3} + \frac{x^5}{!5}\ldots\ldots\right)}{x^5}$$

$$f(0) = \lim_{x\to 0} \frac{x(3+2a+B) + x^3\left(-\frac{3^3}{!3} - \frac{2^3 A}{!3} - \frac{B}{!3}\right) + x^5\left(\frac{3^5}{!5} + \frac{{}^5A}{!5} + \frac{B}{5}\right)\ldots\ldots}{x^5}$$

here

$3 + 2A + B = 0$ So $A = -4$

$\left(\frac{-27}{!3} - \frac{8A}{!3} - \frac{B}{!5}\right) = 0$ $B = 5$

$$f(0) = \lim_{x\to 0} x^5 \frac{\left(\frac{243}{!5} + \frac{32A}{!5} + \frac{B}{!5}\right) + \ldots\ldots}{x^5}$$

$f(0) = 1$

Q9. Let $f(x) = \begin{cases} \frac{\left(\frac{\pi}{2} - \sin^{-1}\left(1-\{x\}^2\right)\right)\sin^{-1}(1-\{x\})}{\sqrt{2}\left(\{x\}-\{x\}^3\right)} & for x \neq 0 \\ \frac{\pi}{2} & for x = 0 \end{cases}$ where {x} is the fractional part of x.

Consider another function g(x) ; such that

$g(x) = f(x)$ for $x \geq 0$

$= 2\sqrt{2}\, f(x)$ for $x < 0$ Discuss the continuity of the functions f(x) & g(x) at x = 0.

Ans. $f(0^+) = \frac{\pi}{2}$; $f(0^-) = \frac{\pi}{4\sqrt{2}}$ ⇒ f is discont. at x = 0 ; $g(0^+) = g(0^-) = g(0) = \pi/2$ ⇒ g is cont. at x = 0

Sol.
$$\lim_{x\to 0^-} f(x) = \lim_{h\to 0} \frac{\left(\frac{\pi}{2} - sin^{-1}(1-\{h\}^2)\right).\, sin^{-1}(1-\{h\})}{\sqrt{2}(\{h\}-\{h\}^3)}$$

$$= \lim_{h\to 0} \frac{\left(\frac{\pi}{2} - sin^{-1}(1-h^2))\right) sin^{-1}(1-h)}{\sqrt{2}(h-h^3)}$$

$$= \lim_{h \to 0} \frac{cos^{-1}(1-h^2)}{\sqrt{2}(1-h^2)} \times \frac{sin^{-1}(1-h)}{h} = \frac{\pi}{2}$$

$$\lim_{x \to 0^-} f(x) = \lim_{h \to 0} \frac{\left(\frac{\pi}{2} - sin^{-1}(1-(h)^2\right) sin^{-1}(1-(1-h)}{\sqrt{2}((1-h)-(1-h)^3)}$$

$$= \lim_{h \to 0} \frac{\frac{\pi}{2} sin^{-1} h}{\sqrt{2}(1-h)(2-h)h} = \frac{\pi}{4\sqrt{2}}$$

Now g(x) = $\begin{cases} \frac{\pi}{2} & ; \ x \geq 0 \\ \\ 2\sqrt{2}\frac{\pi}{4\sqrt{2}} & ; \ x < 0 \end{cases}$

g(x) = $\begin{cases} \frac{\pi}{2} & ; \ x \geq 0 \\ \\ \frac{\pi}{2} & ; \ x < 0 \end{cases}$

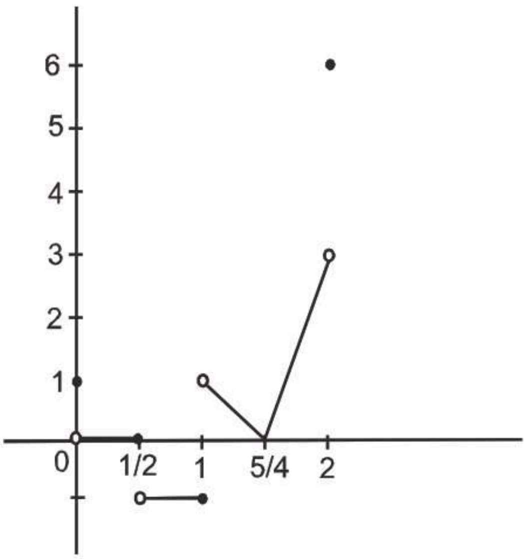
so f(x) is discontinuous at x = 0

Q10. Discuss the continuity of f in [0,2] where f(x) = $\begin{bmatrix} |4x-5|[x] & for x > 1 \\ [\cos \pi x] & for x \leq 1 \end{bmatrix}$; where [x] is the greatest integer not greater than x. Also draw the graph.

Ans. the function f is continuous everywhere in [0 , 2] except for x = 0 ,$\frac{1}{2}$, 1 & 2.

Sol.

$$f(x) = \begin{cases} 1 & , & x = 0 \\ 0 & , & 0 < x \leq 1/2 \\ -1 & , & 1/2 < x \leq 1 \\ 5-4x & , & 1 < x < 5/4 \\ 4x-5 & , & 5/4 \leq x < 2 \\ 6 & , & x = 2 \end{cases}$$



is discontinous at x = 0, $\frac{1}{2}$, 1, 2 in [0, 2]

Q11. If f(x) = x + {–x} + [x] , where [x] is the integral part & {x} is the fractional part of x. Discuss the continuity of f in [– 2, 2].

Ans. discontinuous at all integral values in [– 2 , 2]

Sol. f(x) = x + {–x} + [x]

$$f(x) = \begin{cases} 4 & ; x = 2 \\ x+2-x+1 & ; 2 < x < 1 \\ 1+0+1 & ; x = 1 \\ x+1-x & ; 0 < x < 1 \\ 0 & ; x = 0 \\ x + x - 1 & ; -1 < x < 0 \\ -1 - 1 & ; x = -1 \\ x - x - 1 - 2 & ; -2 < x < -1 \\ -4 & ; x = -2 \end{cases}$$

$$f(x) = \begin{cases} 4 & ; x = 2 \\ 3 & ; 2 < x < 1 \\ 2 & ; x = 1 \\ 1 & ; 0 < x < 1 \\ 0 & ; x = 0 \\ -1 & ; -1 < x < 0 \\ -2 & ; x = -1 \\ 2 & ; -2 < x < -1 \\ -3 & ; x = -2 \end{cases}$$

hence point of discontinuity of x = 2, 1, 0, –1, –2

Q12. Find the locus of (a, b) for which the function $f(x) = \begin{cases} ax - b & for \quad x \le 1 \\ 3x & for \quad 1 < x < 2 \\ bx^2 - a & for \quad x \ge 2 \end{cases}$

is continuous at x = 1 but discontinuous at x = 2.

Ans. locus (a, b) → x, y is y = x – 3 excluding the points where y = 3 intersects it

Sol. L.H.L. = $\lim\limits_{h \to \infty} a(1-h) - b = a - b$

f(1) = a – b

R.H.L. = 3(1 + h) = 3

L.H.L. =f(1) = R.H.L.

a – b = a – b = 3

a – b = 3 .... (1)

but at x = 2

L.H.L $\neq$ f(2)

$\lim_{h\to\infty} 3(2-h) \neq 4b - a$

$6 \neq 4b - a$ .... (2)

common points as (1) and (2)

a = 2 & b = 6

so all (a, b) sotisfing a – b = 3

except a = 3 & b = 6.

Q13. A function f : R → R is defined as $f(x) = \lim_{n\to\infty} \frac{ax^2 + bx + c + e^{nx}}{1 + c.e^{nx}}$ where f is continuous on R. Find the values of a, b and c.

Ans. $c = 1,\ a, b \in R$

Sol. This is built in Limit problem

$$f(x) = \lim_{x\to\infty} \frac{a^2 + bx + c + (e^x)^n}{1 + c(e^x)^n}$$

$$f(x) = \begin{cases} \dfrac{\frac{ax^2}{(e^x)^n} + \frac{bx}{(e^x)^n} + \frac{c}{(e^x)^n} + 1}{\frac{1}{(e^x)^n} + c} & ;\ e^x > 1 \\[2ex] \dfrac{ax^2 + bx + c + 1}{1 + c} & ;\ e^x = 1 \\[2ex] \dfrac{ax^2 + bx + c}{1} & ;\ 0 < e^x < 1 \end{cases}$$

$$f(x) = \begin{cases} \dfrac{1}{c} & ;\ x \in (0, \infty) \\[2ex] 1 & ;\ x = 0 \\[2ex] \dfrac{ax^2 + bx + c}{1} & ;\ x \in (-\infty, 0) \end{cases}$$

Only doutful points is x= 0 L.H.L. = f(0) = R.H.L.

$\frac{1}{e} = 1 \Rightarrow c = 1$ then a, b $\in$ R

Q14. Let f (x) = $x^3 - x^2 - 3x - 1$ and h (x) = $\frac{f(x)}{g(x)}$ where h is a rational function such that (a) it is continuous every where except when x = – 1, (b) $\lim_{x\to\infty} h(x) = \infty$ and (c) $\lim_{x\to -1} h(x) = \frac{1}{2}$.

Find $\lim_{x\to 0}$ (3h(x)+f(x)–2g(x))

Ans. g (x) = 4 (x + 1) and limit = $-\frac{39}{4}$

Sol. $f(x) = x^3 - x^2 - 3x - 1$

$f(x) = (x + 1)(x^2 - 2x - 1)$

Let g(x) = A(x + 1)

$\lim_{x\to -1} h(x) = \frac{1}{2}$ x = –1 only point of discontinuity

$$\lim_{x\to -1} h(x)\frac{f(x)}{g(x)} = \frac{1}{2}$$

$$\lim_{x\to -1} h(x)\frac{(x+)(x^2-2x-1)}{A(x+1)} = \frac{1}{2}$$

A = 4

hence g(x) = 4(x +1)

$$\lim_{x\to 0}(3h(x) + f(x) - 2g(x)$$

$$= \left(-3\times\frac{1}{4} - 1 - 2\times 4\right)$$

$$= \frac{-39}{4}$$

Q15. Find the number of points of discontinuity of the function f(x) = [5x] + {3x} in [0, 5]. where [y] and {y} denote largest integer less than or equal to y and fractional part of y respectively.

Ans. 30

Sol. f (x) = [5x] – [3x] + 3x

Now in [0, 1] we have to check at x = $\frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}, \frac{1}{3}, \frac{2}{3}$

and f (x) is discontinuous at x = $\frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}, \frac{1}{3}, \frac{2}{3}$ i.e. 6 points

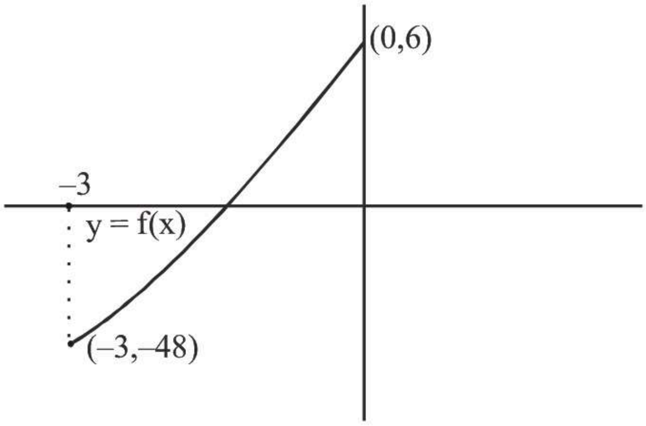
∴ Total points 6 × 5 = 30 **Ans.**

Note : f(x) is continuous at all integers in [0, 5].

Q16. Let f(x) = $x^3 - 3x^2 + 6$ and g(x) = $\begin{cases} \max\{f(t), x+1 \le t \le x+2 & , \quad -3 \le x < 0 \\ 1 - x & , \quad x \ge 0 \end{cases}$. test the continuity of g(x) for x ∈ [–3, 1]

Ans. Discont. at x = 0

Sol. $f(x) = x^3 - 3x^2 + 6$



Q17. Let $f(x) = \begin{cases} \frac{\ln \cos x}{\sqrt[4]{1+x^2}-1} & if \quad x > 0 \\ \frac{e^{\sin 4x}-1}{\ln(1+tan2x)} & if \quad x < 0 \end{cases}$

Is it possible to define f(0) to make the function continuous at x = 0. If yes what is the value of f(0), if not then indicate the nature of discontinuity.

Ans. $f(0^+) = -2$; $f(0^-) = 2$ hence f(0) not possible to define

Sol. $$R.H.L. = \frac{ln\, cosh}{\sqrt[4]{1+h^2}-1}$$

$$R.H.L. = \frac{ln(1+cosh-1)}{\sqrt[4]{1+h^2}-1}\left(\sqrt[4]{1+h^2}+1\right)$$

$$R.H.L. = \frac{ln(1+cosh-1)}{h^2}\left(\sqrt{1+h^2}+1\right)\left(\sqrt[4]{1+h^2}+1\right)$$

$$R.H.L. = \frac{ln(1+cosh-1)}{h^2}.\frac{cosh-1}{h^2}\left(\sqrt{1+h^2}+1\right)\left(\sqrt[4]{1+h^2}+1\right)$$

$$R.H.L. = (1)\left(-\frac{1}{2}\right)(4) = -2$$

$$L.H.L. = \frac{e^{sin4x}-1 \quad sin4x}{\frac{ln(1+tna2x)}{tan2x}\times tan2x\times sin4x}$$

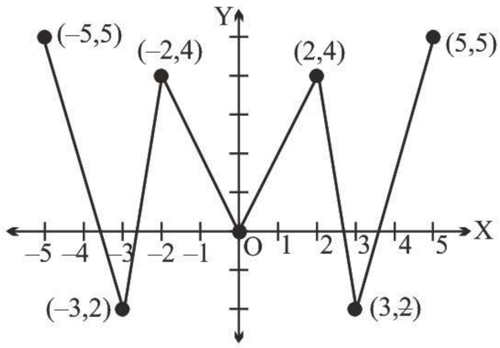
$$L.H.L. = \frac{e^{sin4x}-1}{sin4x}\times\frac{sin4x}{tan2x} = 2$$

hence function is not continuous at x = 0.

Q18. Let f be a real valued continuous function on R and satisfying f (–x) – f (x) = 0 ∀ x ∈ R. If f (– 5) = 5, f (– 2) = 4, f (3) = – 2 and f (0) = 0 then find the minimum number of zero's of the equation f(x) = 0.

Ans. 5

Sol.



One of the possible graph of f (x) having minimum number of zeroes.

Clearly f (x) = 0 has atleast 5 solutions.

Q19. If g : [a, b] onto [a, b] is continous show that there is some c $\in$ [a, b] such that g (c) = c.

Ans. -

Sol. -

Q20. Let f be continuous on the interval [0, 1] to R such that f (0) = f (1). Prove that there exists a point c in $\left[0, \frac{1}{2}\right]$ such that f (c) = f $\left(c+\frac{1}{2}\right)$

Ans. -

Sol. Let

$$h(x) = f(x) - f\left(x + \frac{1}{2}\right)$$

$$h(0) = f(0) - f\left(0 + \frac{1}{2}\right)$$

$$\frac{+\,h\left(\frac{1}{2}\right) = f\left(\frac{1}{2}\right) - f(1)}{h(0) + h\left(\frac{1}{2}\right) = f(0) - f(1)}$$

$$h(0) + h\left(\frac{1}{2}\right) = 0$$

hence By IMVT we can say there exists a 'c' in $\left[0, \frac{1}{2}\right]$, where $f(c) = f\left(c + \frac{1}{2}\right)$

Q21. If the function f (x) defined as f (x) = $\begin{cases} -\frac{x^2}{2} & for \quad x \le 0 \\ x^n \sin\frac{1}{x} & for x > 0 \end{cases}$ is continuous but not derivable at x = 0 then find the range of n.

Ans. 0 < n $\le$ 1

Sol. L.H.L. = 0

f(0) = R.H.L.

R.H.L. $= \lim_{h\to 0} h^{n\sin}\frac{1}{h} = 0$

then continuous for n > 0

for differetiability

$$\mathrm{L.H.D.} = \lim_{h\to 0} \frac{f(0+h)-f(0)}{h} = \lim_{h\to 0} \frac{\frac{-h^2-0}{2}}{h} = 0$$

$$\mathrm{R.H.D.} = \lim_{h\to 0} \frac{h^{n\sin\frac{1}{h}} - f(0)}{h} = \lim_{h\to 0} h^{n-1}\sin\left(\frac{1}{h}\right) = 0$$

n > 1

hence continuous but not differentiable then $0 < n \leq 1$

Q22. Examine the origin for continuity & derrivability in the case of the function f defined by $f(x) = x \tan^{-1}(1/x)$, $x \neq 0$ and f(0) = 0.

Ans. conti. but not diff. at x = 0

Sol. Continuity at x = 0

f(0) = 0

$$\mathrm{R.H.L.} = \lim_{h\to 0} h \tan^{-1}\left(\frac{1}{h}\right) = 0$$

$$\mathrm{L.H.L.} = \lim_{h\to 0} (-h)\tan^{-1}\left(\frac{1}{h}\right) = 0$$

hence continuous at x = 0

differentiablity at x = 0

$$\mathrm{R.H.D.} = \lim_{h\to 0} \frac{f(0+h)-f(0)}{h}$$

$$= \lim_{h\to 0} \frac{h\tan^{-1}\left(\frac{1}{h}\right) - 0}{h} = \frac{\pi}{2}$$

$$\mathrm{L.H.D.} = \lim_{h\to 0} \frac{(-h)\tan^{-1}\left(\frac{1}{h}\right) - 0}{h} = -\frac{\pi}{2}$$

hence function not differentiable at x = 0.

Q23. If f(x)=|x – 1|. ([x] – [–x]) , then find $f'(1+)$ & $f'(1^-)$ where [x] denotes greatest integer function.

Ans. f'(1+) = 3 , f'(1–) = -1

Sol. $$f'(1^+) = \lim_{h\to 0} \frac{f(1+h)-f(1)}{h}$$

$$= \lim_{h\to 0} \frac{|1+h-1|.([1+h]-[-1-h]) - 0}{h}$$

$$= \lim_{h\to 0} [1+h] - [-1-h]$$

$$= 1 - (-2)$$

$= 3$

$$f'(1^-) = \lim_{h\to 0} \frac{f(1-h) - f(1)}{-h}$$

$$= \lim_{h\to 0} \frac{|1-h-1|.([1-h]-[-1+h]) - 0}{-h}$$

$$= \lim_{h\to 0} \frac{0 - 1(-1)}{-1}$$

$$f'(1^-) = -1$$

Q24. If f(x) $= \begin{bmatrix} ax^2 - b & if & |x| < 1 \\ -\frac{1}{|x|} & if & |x| \geq 1 \end{bmatrix}$ is derivable at x = 1. Find the values of a & b.

Ans. a= 1/2 , b = 3/2

Sol.

$$f(x) = \begin{bmatrix} ax^2 - b & ; & -1 < x < 1 \\ \frac{-1}{x} & ; & x \geq 1 \\ \frac{1}{x} & ; & x \leq -1 \end{bmatrix}$$

Now f(x) is differentiable at x = 1

$\Rightarrow 2ax = \frac{1}{x^2}$ at x = 1

$\Rightarrow a = \frac{1}{2}$

Also f(x) is continuous at x = 1

$\Rightarrow a(1)^2 - b = -1 \Rightarrow b = \frac{3}{2}$

Q25. Let f(x) be defined in the interval [–2, 2] such that f(x) = $\begin{bmatrix} -1 & , & - \leq x \leq 0 \\ x - 1 & , & 0 < x \leq 2 \end{bmatrix}$ &

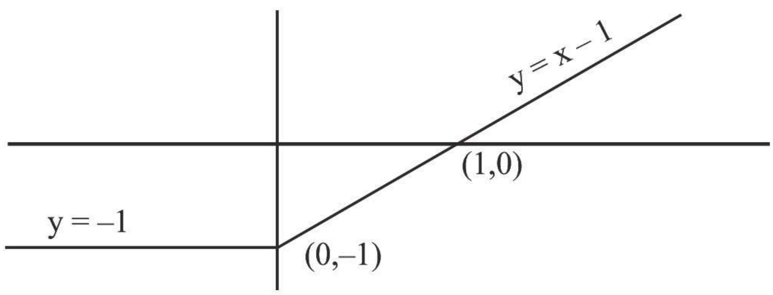
g(x) = f(|x|) + |f(x)|. Test the differentiability of g(x) in (- 2, 2).

Ans. not derivable at x = 0 & x = 1

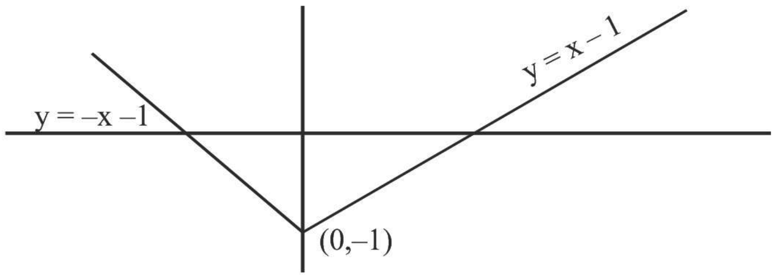
Sol.

$$f(x) = \begin{bmatrix} -1 & ;-2\leq x\leq 0 \\ x-1 & ;\ 0<x\leq 2 \end{bmatrix}$$

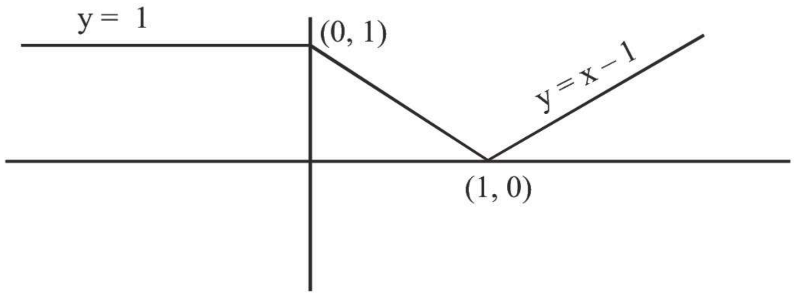
Graph of y = f(x)



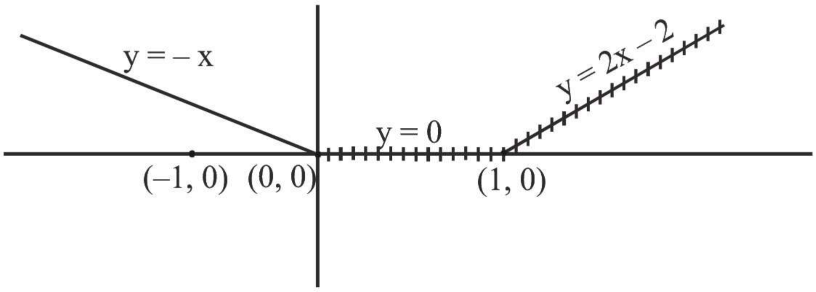
Graph of y = |f(x)|



Graph of y = |f(x)|



Graph of y = f(|x|) + |f(x)|



Not differentiable at x = 0 & 1.

Q26. $f(x) = x.\left(\frac{e^{[x]+|x|}-2}{[x]+|x|}\right)$, $x \neq 0$ & f(0) = –1 where [x] denotes greatest integer less than or equal to x. Test the differentiability of f(x) at x = 0.

Ans. not derivable at x = 0

Sol.

$$R.H.D. = \lim_{h\to 0}\frac{h\left(\frac{e^{[h]+|h|}-2}{[h]+|h|}\right)+1}{h}$$

$$= \lim_{h\to 0}\frac{e^{h}-2+1}{h} = 1$$

$$L.H.D. = \lim_{h\to 0}\frac{-h\left(\frac{e^{[-h]+|-h|}-2}{[-h]+|-h|}\right)+1}{-h}$$

$$= \lim_{h\to 0}\frac{f\frac{-h\left(e^{-1+h}-2\right)}{-1+h}+1}{-h}$$

$= \lim_{h \to 0} \frac{-he^{h-1} + 2h + h - 1}{(h - 1(-h)}$

= DNE

hence f(x) is not derivable at x = 0.

Q27. Discuss the continuity & the derivability in [0 , 2] of f(x) = $\begin{bmatrix} |2x-3|[x] & for & x \geq 1 \\ \sin\frac{\pi x}{2} & for & x < 1 \end{bmatrix}$

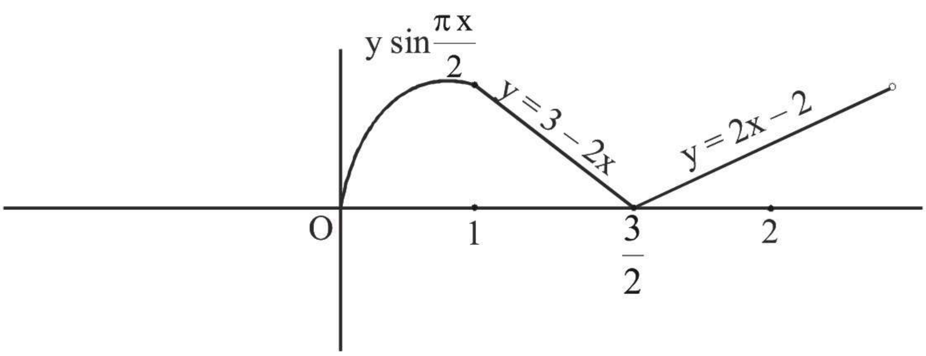
where [ ] denote greatest integer function .

Ans. f is conti. at x = 1 , 3/2 & disconti. at x = 2, f is not diff. at x =1, 3/2 , 2

Sol.

$$f(x) = \begin{cases} 2 & ; \ x = 2 \\ (2x-3)(1) & ; \ \frac{3}{2} < x < 2 \\ (3-2x)(1) & ; \ 1 \leq x < \frac{3}{2} \\ \sin\frac{\pi x}{2} & ; \ 0 \leq x < 1 \end{cases}$$

Graph of y = f(x) (2, 2)



hence points of non-differentiability

$x = 1, \frac{3}{2}, 2$

but not continuous at x = 2.

Q28. Let f (x) = [3 + 4 sin x] (where [ ] denotes the greatest integer function). If sum of all the values of 'x'

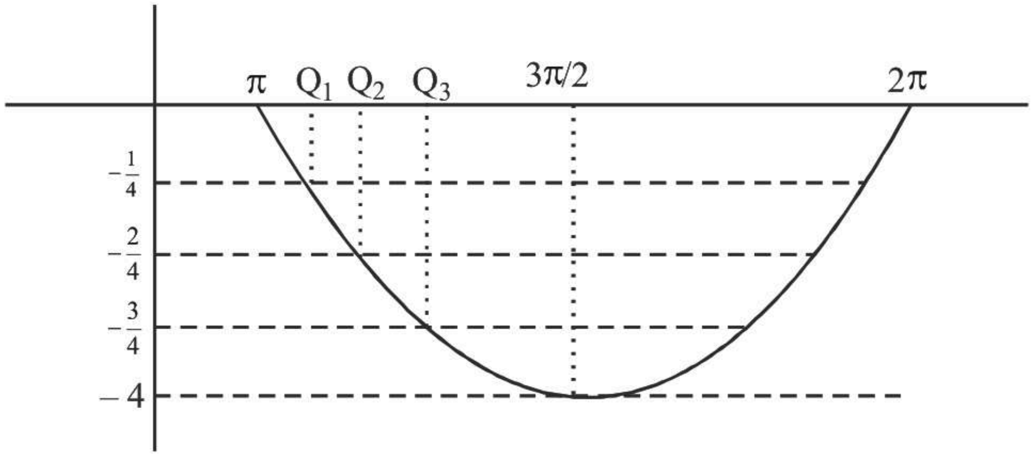
in $[\pi, 2\pi]$ where f (x) fails to be differentiable, is $\frac{k\pi}{2}$, then find the value of k.

Ans. 24

Sol. f(x) = [3 + 4 sin x] $x \in [\pi, 2\pi]$

f(x) = 3 + [4 sin x]

Greatust integer function is doubtful at integer values hence graph of y =[4 sinx]



It is clear function is not differentiable at x = π

$= \pi + sin^{-1}\left(-\frac{1}{4}\right)$

$= \pi + sin^{-1}\left(-\frac{2}{4}\right)$

$= \pi + sin^{-1}\left(-\frac{3}{4}\right)$

$= 2\pi - sin^{-1}\left(-\frac{3}{4}\right)$

$= 2\pi - sin^{-1}\left(-\frac{2}{4}\right)$

$= 2\pi - sin^{-1}\left(-\frac{1}{4}\right)$

$= 2\pi$

sum = 12π

$= \frac{24\pi}{2}$

$= 12\pi$

Q29. The function f(x) = $\begin{cases} ax(x-1)+b & when x & <1 \\ x-1 & when & 1 \le x \le 3 \\ px^2+qx+2 & when & x>3 \end{cases}$

Find the values of the constants a, b, p, q so that

(i) f(x) is continuous for all x (ii) f ' (1) does not exist (iii) f '(x) is continuous at x = 3

Ans. $a \neq 1, b = 0, p = \frac{1}{3}$ and q = – 1

Sol. (i) Continuity at x = 1

L.H.L. = f(1)

0 + b = 1 – 1

b = 0

continuity at x = 3

f(3) = R.H.L.

2 = 9p + 3q + 2

3p + q = 0

$$\text{(ii)}\ f'(x) = \begin{cases} 2ax - a & ;\ x < 1 \\ 1 & ;\ 1 \le x \le 3 \\ 2px+q & ;\ 1 > 3 \end{cases}$$

at x = 1

L.H.D. $\neq$ R.H.D

$2a(1 - h) - a \neq 1$

$2a - a \neq 1$

$a \neq 1$

(iii) Continuity at x = 3 of

L.H.D. = R.H.D.

1 = 6p + q

by (i), (ii), (iii)

$a \neq 1, b = 0, p = \frac{1}{3}, q = -1$

Q30. Examine the function , $f(x) = x.\dfrac{a^{1/x} - a^{-1/x}}{a^{1/x} + a^{-1/x}}$, $x \neq 0$ $(a > 0)$ and $f(0) = 0$ for continuity and existence of the derivative at the origin.

Ans. If $a \in (0, 1)$ $f'(0^+) = -1$; $f'(0^-) = 1$ $\Rightarrow$ continuous but not derivable

$a = 1$; $f(x) = 0$ which is constant $\Rightarrow$ continuous and derivable

If $a > 1$ $f'(0^-) = -1$; $f'(0^+) = 1 \Rightarrow$ continuous but not derivable

Sol. $$R.H.D. = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h\to 0} \frac{h\left(\frac{a^{\frac{1}{h}} - a^{\frac{1}{h}}}{a^{\frac{1}{h}} + a^{\frac{1}{h}}}\right) - 0}{h}$$

$$= \lim_{h\to 0} \left(\frac{1 - a^{\frac{2}{h}}}{1 + a^{\frac{2}{h}}}\right) = 1$$

Case -I   a > 1

P.H.D. $=\left(\frac{1-0}{1+0}\right)=1$

Case-II   a = 1

R.H.D. = 0

Case-III   0 < a < 1

$$R.H.D. = \lim_{h\to 0}\left(\frac{a^{\frac{2}{h}}-1}{a^{\frac{2}{h}}+1}\right)=-1$$

$$L.H.D. = \lim_{h\to 0}\frac{f(0-h)-f(0)}{-h}$$

$$= \lim_{h\to 0}\frac{-h\left(\frac{a^{\frac{1}{h}}-a^{\frac{1}{h}}}{a^{\frac{1}{h}}+a^{\frac{1}{h}}}\right)-0}{-h}$$

Case - I   a > 1

$$L.H.D. = \lim_{h\to 0}\left(\frac{1-a^{\frac{2}{h}}}{1-a^{\frac{2}{h}}}\right)=-1$$

Case - II   a = 1
L.H.D. = 0

Case-III   0 < 1 < 1

L.H.D. = 1

hence

if   0 < a < 1

L.H.D. = 1   Not differentiable but continuous.

R.H.D. = –1

if   a = 1

L.H.D. = R.H.D.   hence differentiable and continuous

if   a > 1

L.H.D. = –1   Not differentiable but continuous

R.H.D. = 1

Q31. Discuss the continuity & the derivability of 'f' where f (x) = degree of ($u^{x^2}+u^2+2u-3$) at x = $\sqrt{2}$.

Ans. continuous but not derivable at x = $\sqrt{2}$

Sol. f(x) = Degree of $\left(4x^2+4^2+24-3\right)$

$$f(x) = \begin{bmatrix} x^2 & ; x \le \sqrt{2} \\ \\ 2 & ; x \le \sqrt{2} \end{bmatrix}$$

at $x = \sqrt{2}$

L.H.L. = R.H.L. = $f(\sqrt{2}) = 2$

hence continuous at x = $\sqrt{2}$

By Quick Concepts

$$f(x) = \begin{bmatrix} 2x & ; x > \sqrt{2} \\ \\ 0 & ; x \le \sqrt{2} \end{bmatrix}$$

Clearly not at $x \neq 1$ differentiable at x = $\sqrt{2}$.

Q32. If $f(x) = \begin{cases} \frac{\sin[|x|]\pi}{ax^4} + ax^2 + b; & -2 \le x < -1 \\ 3\sec \pi x + \cot^{-1} x; & -1 \le x \le \frac{-3}{4} \end{cases}$
is differentiable in $\left(-2, \frac{-3}{4}\right)$ then $a = \frac{1}{\lambda_1}$ and $b = \frac{3\pi - \lambda_2}{4}$. Find the value of $(\lambda_1 + \lambda_2)$.

Note: [y] denotes largest integer $\le$ y.

Ans. 17

Sol. Clearly $f(x) = \begin{cases} ax^2+b & ; -2 \le x < -1 \\ \\ 3\sec\pi x + \cot^{-1} x & ; -1 \le x \le \frac{-3}{4} \end{cases}$

As the function is differentiable in $\left(-2, \frac{-3}{4}\right)$, so function is differentiable at x = –1.

$\therefore f'(-1-) = f'(-1+)$ .....(1)

$$f'(-1^-) = \lim_{h\to 0} \frac{f(-1-h) - f(-1)}{-h} = \lim_{h\to 0} \frac{\left(a(-1-h)^2 + b\right) - (a+b)}{-h}$$

$$= \lim_{h\to 0} \frac{a(1+h^2-2h) - a}{-h} = \lim_{h\to 0} \frac{ah(h-2)}{-h} = -2a \quad .....(2)$$

$$f'(-1^+) = \lim_{h\to 0} \frac{f(-1+h) - f(-1)}{h} = \lim_{h\to 0} \frac{3sec\pi(-1+h) + cot^{-1}(-1+h) - 3sec(-\pi) - cot^{-1}(-1)}{h}$$

$$= \lim_{h\to 0} \frac{-3sec\pi h + cot^{-1}(-1+h) + 3 - \frac{3\pi}{4}}{h} = \lim_{h\to 0} \frac{3(1 - sec\pi h)}{h} + \lim_{h\to 0} \frac{cot^{-1}(-1+h) - \frac{3\pi}{4}}{h}$$

$$= \lim_{h\to 0} \frac{3(cos\pi - h - 1)}{hcos\pi h} + \lim_{\theta\to \frac{3\pi^-}{4}} \frac{\theta - \frac{3\pi}{4}}{1 + cot\theta} \quad (\text{Let } \cot^{-1}(-1+h) = \theta)$$

$$= (3\times 0) + \lim_{\theta\to 0^-} \frac{\alpha}{1 + cot\left(\frac{2\pi}{4} + \alpha\right)} \quad \left(\text{Put } \theta = \frac{3\pi}{4} + \alpha\right)$$

$$= 0 + \lim_{\alpha\to 0^-} = 0 + \lim_{\alpha\to 0^-} \frac{\alpha}{1 - \left(\frac{1+tan\alpha}{1-tan\alpha}\right)} = 0 + \lim_{\alpha\to 0^-} \frac{\alpha(1 - tan\alpha)}{(-2tan\alpha)} = \frac{-1}{2}$$

$\therefore$ Using equation (2) and equation (3) in equation (1), we get

$$-2a = \frac{-1}{2} \Rightarrow a = \frac{1}{4} = \frac{1}{\lambda_1} \qquad \text{(Given)}$$

$$\therefore \lambda_1 = 4$$

Also function will be continuous at x = – 1.

$$\therefore f(x) = \lim_{x \to -1^{-1}} f(x) = \lim_{x \to -1^{+1}} f(x)$$

$$\Rightarrow a + b = -3 + \frac{3\pi}{4}$$

$$\therefore b = \frac{-1}{4} - 3 + \frac{3\pi}{4} = \frac{3\pi - 13}{4} = \frac{3\pi - \lambda_2}{4}$$

$$\Rightarrow \lambda_2 = 13$$

Hence $(\lambda_1 + \lambda_2) = 4 + 13 = 17$

Q33. Let $f(x) = \left[\begin{array}{ll} a\cos^{-1}\left(\frac{b+x}{4}\right), & \frac{-2}{3} < x < 0 \\ 2, & x = 0 \\ \frac{ln(1-cx)}{x} & 0 < x < \frac{2}{3} \end{array}\right.$

If the function f (x) is differentiable at x = 0, then find the value of $(b^2 - 2a + c^6)$.

Ans. 48

Sol. As f (x) is derivable at x = 0, so f (x) is also continuous at x = 0.

$$\therefore f(0^+) \lim_{h \to 0} \frac{ln(1-ch)}{h}\left(\frac{0}{0}\right) = \lim_{h \to 0} \frac{-c \times ln(1-ch)}{-ch} = -c$$

$$\Rightarrow -c = 2 \Rightarrow c = -2 \qquad \text{...........(1)}$$

$$\text{Now } f'(0^+) = \lim_{h \to 0} \frac{\frac{ln(1+2h)}{h} - 2}{h^2} = \lim_{h \to 0} \frac{\left(2h - \frac{(2h)^2}{2} + \ldots\ldots\right) - 2h}{h^2} = -2$$

$$\text{Also } f'(0^-) = \lim_{h \to 0} \frac{acot^{-1}\left(\frac{b-h}{4}\right) - 2}{-h}\left(\frac{0}{0}\right) = \lim_{h \to 0} \frac{\frac{-a}{1+\left(\frac{b-h}{4}\right)} \times \frac{-1}{4}}{-1} = \frac{-4a}{b^2 + 16}$$

As $f'(0^-) = f'(0^+)$, so $\frac{-4a}{b^2+16} = -2$

$$\Rightarrow 2a = b^2 + 16$$

$$\therefore n^2 - 2a = 16 \qquad \text{.........(2)}$$

Hence $(b^2 - 2a + c^6) = -16 + 64 = 48$ (Using equation (1) and equation (2))

Q34. If $|f(x_1) - f(x_2)| \le (x_1 - x_2)^2$, for all $x_1, x_2 \in R$. Find the equation of tangent to the curve y = f (x) at the point (1, 2).

Ans. $y - 2 = 0$

Sol. Put $x_1 = x + h$ & $x_2 = x$

$|f(x + h) - f(x)| \leq h^2$

$$\lim_{h\to 0} \frac{f(x+h)-f(x)}{h}\Big| \leq \lim_{h\to 0} h$$

$|f'(x)| \leq 0$

Possibible only if $f'(x) = 0$

$f(x) = c$

at point (1, 2) $f(x) = 2$

$y = 2$

Q35. Let f be a function that is differentiable every where and that has the following properties:

(i) $f(x + h) = f(x) \cdot f(h)$ (ii) $f(x) > 0$ for all real x. (iii) $f'(0) = -1$

Use the definition of derivative to find $f'(x)$ in terms of $f(x)$.

Ans. $f'(x) = -f(x)$

Sol.
$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$
$$f'(0) = \frac{f(0+h)-f(0)}{h} = -1$$
$$f'(x) = \lim_{h\to 0} \frac{f(x)f(h)-f(x)}{h}$$
$$f'(x) = f(x)\Big(\lim_{h\to 0} \frac{f(h)-1)}{h}\Big)$$

We are given

$f(x + h) = f(x)\, f(h)$

$x = h = 0$

$f(0) = f^2(0)$

$f(0) = 0, 1$

$f(0) \neq 0$

$f(0) = 1$

$$f'(x) = f(x)\Big(\lim_{h\to 0} \frac{f(h)-f(0)}{h}\Big)$$

$f'(x) = f(x)\,(-1)$

Q36. A derivable function $f : R^+ \to R$ satisfies the condition $f(x) - f(y) \geq \ln + x - y$ for every $x, y \in R^+$. If g denotes the derivative of f then compute the value of the sum

$$\sum_{n=1}^{100} g\left(\frac{1}{n}\right).$$

Ans. 5150

Sol. $$f'(x) = \lim_{h\to\infty} \frac{f(x+h)-f(x)}{h}$$

$$\geq \lim_{h\to 0} \frac{ln\left(\frac{x+h}{x}\right)+x+h-x}{h}$$

$$\geq \lim_{h\to 0} \frac{ln\left(1+\frac{h}{x}\right)}{h}+1 \geq \frac{1}{x}+1 \qquad \text{..........(i)}$$

$$f'(x) = \lim_{h\to 0} \frac{f(x-h)-f(x)}{-h}$$

$$\leq \lim_{h\to 0} \frac{ln\left(\frac{x-h}{x}\right)+x-h-x}{-h}$$

$$\leq \lim_{h\to 0} \frac{ln\left(1-\frac{h}{x}\right)}{-h}+1 \leq \frac{1}{x}+1 \qquad \text{..........(ii)}$$

from (i) and (ii)

$$\Rightarrow f'(x) = \frac{1}{x}+1$$

\(\displaystyle\sum_{i=1}^{10} =g\Bigg(\frac{1}{n}\Bigg)+g\Bigg(\frac{1}{2}\Bigg)+....+g \Bigg(\frac{1}{100}\Bigg

= (1 + 2 + 3 + ... 100) + 100 = 5150

Paragraph

Consider a function f(x) in $[0, 2\pi]$ defined as :

\n f(x) = $\begin{cases} \backslash n[\sin x]+[\cos x] & ; \quad 0\leq x\leq \pi \\ \backslash n[\sin x]-[\cos x] & ; \quad \pi < x \leq \backslash n 2\pi \end{cases}$

\n where [.] denotes greatest integer function then

\n

Q37. Number of points where f(x) is non-derivable :

(A) 2 (B) 3 (C) 4 (D) 5

Ans. C

Sol.

$$f(x) = \begin{cases} 1 & ; x = 0 \\ 0 & ; 0 < x < \dfrac{\pi}{2} \\ 1 & ; x = \dfrac{\pi}{2} \\ -1 & ; \dfrac{\pi}{2} < x < \pi \\ -1 & ; x = \pi \\ -2 & ; \pi < x < \dfrac{3\pi}{2} \\ -1 & ; x = \dfrac{3\pi}{2} \\ -1 & ; \dfrac{3\pi}{2} < x < 2\pi \\ -1 & ; x = 2\pi \end{cases}$$

Clearly function is non-derivable at $x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$

Paragraph

Consider a function f(x) in $[0, 2\pi]$ defined as :

\n $\quad f(x) = \begin{cases} \backslash n[\sin x] + [\cos x] & ; \quad 0 \le x \le \pi \\ \backslash n[\sin x] - [\cos x] & ; \quad \pi < x \le \backslash n 2\pi \end{cases}$

\n where [.] denotes greatest integer function then

\n

Q38. $\lim\limits_{x \to \left(\frac{3\pi}{2}\right)^+}$ equals

(A) 0 (B) 1 (C) -1 (D) 2

Ans. C

Sol. $\lim\limits_{x \to \infty} \left(\frac{3\pi}{2}\right)^+ f(x) = -1$

Paragraph

Let $f(x) = \begin{cases} \backslash n x[x] & 0 \le x < 2 \\ \backslash n(x-1)[x] & 2 \le x \le 3 \backslash n \end{cases}$ where [x] = greatest integer less than or equal to x, then

\n

Q39. The number of values of x for $x \in [0, 3]$ where f(x) is discontinuous is

(A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol.

$$f(x) = \begin{cases} 0 & ; \ 0 \le x < 1 \\ x & ; 1 \le x < 2 \\ 2x - 2 & ; 2 \le x < 3 \\ 6 & ; x = 3 \end{cases}$$

Clearly function is not continuous at = 1, 3

option (c) .

Paragraph

Let $f(x) = \begin{cases} \backslash n x[x] & 0 \le x < 2 \\ \backslash n(x-1)[x] & 2 \le x \le 3 \backslash n \end{cases}$ where [x] = greatest integer less than or equal to x, then

\n

Q40. The number of values of x for $x \in [0, 3]$ where f(x) is non-differentiable is :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. D

Sol.

$$f'(x) = \begin{cases} 0 & ; \ 0 \le x < 1 \\ 1 & ; 1 \le x < 2 \\ 2 & ; 2 \le x < 3 \\ 0 & ; x = 3 \end{cases}$$

Not differentiable at x = 1, 2, 3.

Paragraph

Let $f(x) = \begin{cases} \backslash n x[x] & 0 \le x < 2 \\ \backslash n(x-1)[x] & 2 \le x \le 3 \backslash n \end{cases}$ where [x] = greatest integer less than or equal to x, then

\n

Q41. The number of integers in the range of y = f(x) is

(A) 3 (B) 4 (C) 5 (D) 6

Ans. C

Sol. Number of integer $= 0$

$= 1$

$= 4$

$= 6$

$= 3$

hence correct option is (C).

Q42. If $f(x) = \begin{cases} -x - \frac{\pi}{2}, & x \le \frac{\pi}{2} \\ -\cos x, & -\frac{\pi}{2} < x \le 0 \\ x - 1, & 0 < x \le 1 \\ ln\, x & x > 1 \end{cases}$ then :

(A) f(x) is continuous at x = $-\dfrac{\pi}{2}$ (B) f(x) is not differentiable at x = 0

(C) f(x) differentiable at x = 1 (D) f(x) is differentiable at x = $-\dfrac{3}{2}$

Ans. A, B, C, D

Sol.

$$f'(x) = \begin{cases} -1 & x \le \frac{\pi}{2} \\ \sin x & -\frac{\pi}{2} < x \le 0 \\ 1 & 0 < x \le 1 \\ \frac{1}{x} & x > 1 \end{cases}$$

$f'(0-) = 0$, $f'(0+) = 1$ $\therefore$ not differentiable at x = 0

$f'(1-) = 1$, $f'(1+) = 1$ $\therefore$ differentiable at x = 1

as $-\dfrac{3}{2} \in \left(-\dfrac{\pi}{2}, 0\right)$

$f'(x) = \sin x$ which is differentiable at x = $-\dfrac{3}{2}$

Q43. Which one of the following statements is NOT CORRECT :

(A) The derivative of a differentiable periodic function is a periodic function with the same period

(B) If f(x) and g(x) both are defined on the entire number line and are aperiodic then the function F(x) = f(x) . g(x) can not be periodic

(C) Derivative of an even differentiable function is an odd function and derivative of an odd differentiable function is an even function

(D) Every function f(x) can be represented as the sum of an even and an odd function

Ans. B, C

Sol. (A) $f(x+T) = f(x) \forall x \in R$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$\Rightarrow f'(x+T) = \lim_{h \to 0} \frac{f(x+h+T) - f(x+T)}{h}$$

= f ′ (x)

(B) $\left.\begin{array}{c} f(x)=e^{x} \\ g(x)=e^{-x} \end{array}\right] \rightarrow aperiodic$

$\Rightarrow f(x)\ (x)=1 \rightarrow Periodic$

(C) f(–x) = f(x)

$$f\prime(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

$$f\prime(-x)=\lim_{h\to 0}\frac{f(-x+h)-f(-x)}{h}$$

$$=\lim_{h\to 0}\frac{f(x-h)-f(x)}{h}$$

= – f ′ (x)

(D) Domain should be symmetric about origin.

Take f(x) = $\log_e$x.

Q44. If f(x) = $\begin{cases} \frac{x\cdot\ell n(\cos x)}{\ell n(1+x^2)} & x\neq 0 \\ 0 & x=0 \end{cases}$ then :

(A) f is continuous at x = 0

(B) f is continuous at x = 0 but not differentiable at x = 0

(C) f is differentiable at x = 0 (D) f is not continuous at x = 0 .

Ans. A, C

Sol.

$$f'(0^+)=\lim_{h\to 0}\frac{hln(cos\,h)}{hln(1+h^2)}=\lim_{h\to 0}\frac{\frac{ln(cosh)^{\frac{1}{h}^2}}{ln(1+h^2)}}{h^2}=\lim_{h\to 0}\frac{1}{h^2}(cosh-1)=-\frac{1}{2};|||1y\ f'(0^{-1})=-\frac{1}{2}$$

hence f is continuous and derivable at x = 0

Q45. Let f (x) be defined as follows: f (x) =$\begin{cases} x^6, & x^2>1 \\ x^3, & x^2\leq 1 \end{cases}$. Then f (x) is:

(A) continuous everywhere (B) differentiable everywhere

(C) discontinuous at x = - 1 (D) not differentiable at x = 1

Ans. C, D

Sol.

$$f'(x)=\begin{bmatrix} x^6 & ;\ x\ (-\infty,-1)\cup(1,\infty) \\ & \\ x^3 & ;\ x\in[-1,1] \end{bmatrix}$$

Continuity at x = 1 Continuity at x = –1

f(1) = 1 f(–1) = –1

L.H.L. =1 L.H.L. =1

R.H.L. = 1 R.H.L. = –1

hence f(x) is continuous at x = 1 but not continuous at x = –1.

$$f'(x) = \begin{cases} 6x^5 & ; x\ (-\infty,-1)\cup(1,\infty) \\ 3x^2 & ; x \in[-1,1] \end{cases}$$

differentiability

at x = 1 x = –1

L.H.D. = 3 L.H.D. = –6

R.H.D. = 6 R.H.D. = 3.

hence the correct options are C & D.

Q46. Which of the following functions f has/have a removable discontinuity at the indicated point?

(A) $f(x) = \frac{x^2-2x-8}{x+2}$ at x = –2 (B) $f(x) = \frac{x-7}{|x-7|}$ at x = 7

(C) $f(x) = \frac{x^3+64}{x+4}$ at x = –4 (D) $f(x) = \frac{3-\sqrt{x}}{9-x}$ at x = 9

Ans. A, C, D

Sol. (A) $f(x) = \frac{(x-4)(x+2)}{(x+2)}$

at x = –2

R.H.L. = –6 Removable. at x = –2

L.H.L. = –6

at x = 7

(B) $f(x) = \frac{(x-7)}{|x-7|}$

at x = 7

R.H.L. = 1 Non Removable at x = 7.

L.H.L. = –1

(C) $f(x) = \frac{(x+4)(x^2+4x+16)}{x+4}$

R.H.L. = 16

L.H.L = 16 Removable at x= –4

(D) $f(x) = \frac{3-\sqrt{x}}{(3+\sqrt{x})(3-\sqrt{x})}$

at x = 9

$R.H.L. = \frac{1}{6}$ Removable at x = 9.

Correct options are (A), (C), & (D).

Q47. In which of the following cases the given equations has atleast one root in the indicated interval?

(A) $x - \cos x = 0$ in $(0, \pi/2)$ (B) $x + \sin x = 1$ in $(0, \pi/6)$

(C) $\frac{a}{x-1} + \frac{b}{x-3} = 0$, a, b > 0 in (1, 3)

(D) f (x) – g (x) = 0 in (a, b) where f and g are continuous on [a, b] and f (a) > g (a) and f (b) < g (b).

Ans. A, B, C, D

Sol. ere we are using IMVT

(A)

f(x) = x – cosx f(x) is continuous and f(a) & f(b) are of opposite

f(0) = – 1 sign hence f(x) = 0 has a solution.

$f\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$

(B)

x + sinx –1 = 0

f(x) = x + sinx –1

f(0) = –1 = – ve

$f\left(\frac{\pi}{6}\right) = \frac{\pi}{6} = \frac{1}{2} - 1 = -ve$

hence no solution.

(C)

$\frac{a}{x-1} + \frac{b}{x-3} = 0$

a(x – 3) + b(x –1) = 0

f(x) = a(x–3) + b(x –1)

f(1) = a(–2) = –ve

f(3) = 2b = +ve

hence there is solution.

(D)
h(x) = f(x) – g(x)

h(a) = f(a) –g(a) = +ve

h(b) = f(b) –g(b) = –ve

hence there is solution.

Q48. Indicate all correct alternatives if, f (x) = $\frac{x}{2}-1$, then on the interval [0, π]

(A) tan (f (x)) & $\frac{1}{f(x)}$ are both continuous

(B) tan (f (x)) & $\frac{1}{f(x)}$ are both discontinuous

(C) tan (f (x)) & f^{-1} (x) are both continuous

(D) tan (f (x)) is continuous but $\frac{1}{f(x)}$ is not

Ans. C, D

Sol. (i) $tan f(x) = tan\left(\frac{x}{2}-1\right) \quad x \in [0, \pi]$

$0 \leq x \leq \pi \Rightarrow -1 \leq \frac{x}{2} - 1 \leq \frac{\pi}{2} - 1$

By graph we say tan(f(x)) is continuous in [0, π]

(ii) $\frac{1}{f(x)} = \frac{2}{x-2}$ is not defined at $x = 2 \in [0, \pi]$

(iii) $y = \frac{x-2}{2}$

$\Rightarrow f^{-1}(x) = 2x + 2$ in continuous in R.

Q49. Which of the following function(s) not defined at x = 0 has/have non-removable discontinuity at the point x = 0 ?

(A) f(x) = $\frac{1}{1+2^{\frac{1}{x}}}$ (B) f(x) = arc tan $\frac{1}{x}$ (C) f (x) = $\frac{e^{\frac{1}{x}}-1}{e^{\frac{1}{x}}+1}$ (D) f (x) = $\frac{1}{\ell n|x|}$

Ans. A, B, C

Sol. (A) at x = 0

$R.H.L. = \frac{1}{1+2^{\frac{1}{h}}} = 0$ Non removable

$L.H.L. = \frac{1}{1+2^{\frac{1}{-h}}} = 1$

(B) $f(x) = tan^{-1}\left(\frac{1}{x}\right)$

$R.H.L. = tan^{-1}\left(\frac{1}{h}\right) = \frac{\pi}{2}$ Non removable

$L.H.L. = tan^{-1}\left(-\frac{1}{h}\right) = -\frac{\pi}{2}$

(C) $f(x) = \frac{e^{\frac{1}{x}} - 1}{e^{\frac{1}{x}} + 1}$

$R.H.L. = \frac{e^{\frac{1}{h}} - 1}{e^{\frac{1}{h}} + 1} = \frac{1 - e^{\frac{1}{h}}}{1 + e^{\frac{1}{h}}} = 1$

$L.H.L. = \frac{e^{\frac{1}{h}} - 1}{e^{\frac{1}{h}} + 1} = -1$ Non removable

(D) $f(x) = \frac{1}{ln|x|}$

R.H.L. = 0

L.H.L. = 0 Removable.

Q50. The function, f (x) = [|x|] - |[x]| where [x] denotes greatest integer function

(A) is continuous for all positive integers

(B) is discontinuous for all non positive integers

(C) has finite number of elements in its range

(D) is such that its graph does not lie above the x - axis .

Ans. A, B, C, D

Sol. f(x) = [|x|] – |[x]|

By Looking at option we check

at x = 0

R.H.L. = [|h|] – |[h]| = 0 and f(0) = 0

L.H.L. = [|h|] – |–1| = –1

at x = 1

R.H.L. = [|1+ h|] – |[1+ h]| = 0 and f(1) = 0

L.H.L. = [|1– h|] – |[1– h]| = 0

at = 2

R.H.L = [|2+ h|] – |[2+ h]| = 2 – 2 = 0 f(2) = 0

L.H.L. = [|2 –h|] – |[2 –h]| = 1 –1 = 0

at $x = -2$

R.H.L. $= [|-2+h|] - |[-2+h]| = -1$

L.H.L $= [|-2-h|] - |[-2-h]| = 2 - 3 = -1$

$f(-2) = 0$

hence correct answers are (A), (B), (C), (D) .

Q51. Given that the derivative f' (a) exists. Indicate which of the following statement(s) is/are always True

(A) $f'(a) = \lim\limits_{h \to a} \dfrac{f(h) - f(a)}{h - a}$ (B) $f'(a) = \lim\limits_{h \to 0} \dfrac{f(a) - f(a-h)}{h}$

(C) $f'(a) = \lim\limits_{h \to 0} \dfrac{f(a+2t) - f(a)}{t}$ (D) $f'(a) = \lim\limits_{h \to 0} \dfrac{f(a+2t) - f(a+t)}{2t}$

Ans. A, B

Sol. (C) is false and is True only if f ' (a) = 0 limit is 2f ' (a). In (D) same logic limit is $\frac{1}{2}f'(a)$.

Q52. The function $f(x) = \sqrt{1 - \sqrt{1 - x^2}}$

(A) has its domain $-1 \le x \le 1$. (B) has finite one sided derivates at the point x = 0.

(C) is continuous and differentiable at x = 0.

(D) is continuous but not differentiable at x = 0.

Ans. A, B, D

Sol. $f'(0) = \frac{1}{\sqrt{2}}; f'(0^-) = -\frac{1}{\sqrt{2}}; f(x) = \frac{\sqrt{x^2}}{\sqrt{1+\sqrt{1-x^2}}} = \frac{|x|}{\sqrt{1+\sqrt{1-x^2}}}$

Q53. Let $f : (0, \infty) \to [1, \infty)$ be defined as $f(x) = \begin{cases} \dfrac{1}{x}, & 0 < x < 1 \\ \dfrac{x}{[x]}, & 1 \le x < \infty \end{cases}$

where $[x]$ denotes largest integer less than or equal to x. Then which of the following statements is (are) correct ?

(A) $f(x)$ is neither injective nor surjective.

(B) $f(x)$ is continuous but non-derivable at $x = 1$.

(C) $f(x)$ has non-removable discontinuity of finite type at $x = 2$ with jump equals $\frac{1}{2}$.

(D) $f'(e) = \frac{1}{2}$.

Ans. B, D

Sol. (A) Clearly $f\left(\frac{2}{3}\right) = \frac{3}{2}$ and $f\left(\frac{3}{2}\right) = \frac{3}{2}$

So, f (x) is many one function.

Also range of f(x) = $[1, \infty)$, so

f(x) is onto also. $\Rightarrow$ (A) is incorrect]

(B) $f(1^-) = 1 = f(1^+) = f(1)$

But $f'(1-) = -1$ and $f'(1+) = 1$

So, f (x) is continuous but non-derviable at x = 1 $\Rightarrow$ (B) is correct.

(C) $f(2^+) = 1$, $f(2^-) = 2$

$\lim_{x\to 2} f(x)$ does not exist.

Q54. f(x) = |x[x]| in - 1 ≤ x ≤ 2 , where [x] is greatest integer ≤ x then f(x) is :

(A) continuous at x = 0 (B) discontinuous x = 0 (C) not differentiable at x = 2

(D) differentiable at x = 2

Ans. A, C

Sol. f(x) = |x [x]|

$$= \begin{cases} -x & ; -1 \le x < 0 \\ 0 & ; 0 \le x < 1 \\ x & ; 1 < x < 2 \\ 4 & ; x = 2 \end{cases}$$

at x = 0

f(0) = 0 hence continuous at x = 0.

R.H.L. =0

L.H.L. = 0

at x = 2 function is not continuous then it cannot be differentiable at x =2.

hence correct answers are (A) & (C).

Q55. f(x) =1 + x.[cosx] in $0 < x \le \pi/2$, where [ ] denotes greatest integer function then ,

(A) It is continuous in $0 < x < \pi/2$ (B) It is differentiable in $0 < x < \pi/2$

(C) Its maximum value is 2 (D) It is not differentiable in $0 < x < \pi/2$

Ans. B, A

Sol. $f(x) = 1 + x(0); 0 < x \le \frac{\pi}{2}$

$f(x) = 1; 0 < x \le \frac{\pi}{2}$

hence correct answers are (A) & (B)

Q56. Select the correct statements.

(A) The function f defined by $f(x) = \begin{cases} 2x^2 + 3 & for x \le 1 \\ 3x + 2 & for x > 1 \end{cases}$ is neither differentiable nor continuous at $x = 1$.

(B) The function $f(x) = x^2 |x|$ is twice differentiable at $x = 0$.

(C) If f is continuous at x = 5 and f (5) = 2 then $\lim_{x\to 2} f(4x^2 - 11)$ exists.

(D) If $\lim_{x\to a} (f(x) + g(x)) = 2$ and $\lim_{x\to a}(f(x) - g(x)) = 1$ then $\lim_{x\to a} f(x) \cdot g(x)$ need not exist.

Ans. B, C

Sol. (A) Continuity at x = 1

f(1) = L.H.L. = R.H.L. = 5

By Quick concept

$$f'(x) = \begin{cases} 4x & ; x \le 1 \\ 3 & ; x > 1 \end{cases}$$

L.H.D. = 4

R.H.D. = 3

(B)

$$f(x) = \begin{cases} x^3 & ; x \ge 0 \\ -x^3 & ; x < 0 \end{cases}$$

$$f'(x) = \begin{cases} 3x^2 & ; x \ge 0 \\ -3x^2 & ; x < 0 \end{cases}$$

L.H.D. = 0

R.H.D. = 0

$$f''(x) \begin{cases} 6x & ; x \ge 0 \\ -6x & ; x < 0 \end{cases}$$

L.H.D. = 0

R.H.D. = 0

(C) Obviously a correct statement because $\lim_{x\to 2} f(5) = f(5)$

(D) Not necessarily a correct ststement.

Q57. Let f(x) = |2x – 9| + |2x| + |2x + 9|. Which of the following are true ?

(A) f(x) is not differentiable at = $\frac{9}{2}$ (B) f(x) is not differentiable at $\frac{-9}{2}$

(C) f(x) is not differentiable at x = 0 (D) f(x) is differentiable at x = $\frac{-9}{2}, 0, \frac{9}{2}$

Ans. A, B, C

Sol.

$$f(x) = \begin{cases} -6x & ; x \le -\frac{9}{2} \\ -2x + 18 & ; -\frac{9}{2} < x < 0 \\ 2x + 18 & ; 0 \le x \le \frac{9}{2} \\ 6x & ; x \ge \frac{9}{2} \end{cases}$$

Continuity at x = 0

L.H.L. = 18

(0) = 8

R.H.L. = 8

Continuity at x = $\frac{9}{2}$

L.H.L. = R.H.L. = $f\left(-\frac{9}{2}\right)$ = 27

hence continuous everywhere

By Quick Concept

$$f'(x) = \begin{cases} -6 & ; x \le -\frac{9}{2} \\ -2 & ; -\frac{9}{2} < x < 0 \\ 2 & ; 0 \le x \le \frac{9}{2} \\ 6 & ; x \ge \frac{9}{2} \end{cases}$$

hence correct options are (A) , (B), (C).

Q58. Column-I

Column-II

(A) If f (x) is derivable at x = 3 & f '(3) = 2, then $\lim_{h\to 0} \frac{f(3+h^2) - f((3-h^2)}{2h^2}$ equals

(P) 0

(B) Let f(x) be a function satisfying the condition f(–x) = f(x) for all real x. If f' (0)
(Q) 1
exists, then its value is equal to

(C) For the function f(x) = $\begin{bmatrix} \frac{x}{1+e^{1/x}}, & x \neq 0 \\ 0, & x = 0 \end{bmatrix}$, the derivative from the left, f'(0^-) equals
(R) 2

(D) The number of points at which the function f(x) = max. {a – x, a + x, b}, – ∞ < x < ∞, (S) 3
0 < a < b cannot be differentiable is

(A) (A) R, (B) P, (C) Q, (D) R   (B) (A) S, (B) S, (C) Q, (D) R

(C) (A) R, (B) P, (C) S, (D) P   (D) (A) R, (B) S, (C) Q, (D) S

Ans. 5f158bd708c2efb274bf1fcf

Sol. $\lim\limits_{h \to 0} \frac{f(3+h^2) - f(3-h^2)}{(3+h^2)-(3-h^2)}$

= f ' (3) = 2

(B) –f ' (– x) = f ' (x)

x = 0

– f ' (0) = f ' (0)

2f ' (0) = 0

f ' (0) = 0

(C) $\text{L.H.D.} = \lim\limits_{h \to 0} \frac{f(0-h) - f(0)}{-h}$

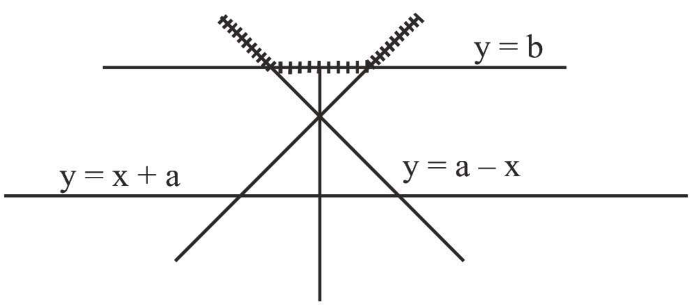
(D) y = a – x

y = a + x

y = b

By Graphical method

Graph



$$y = f(x) = \begin{cases} a - x & ; x \le a - b \\ b & ; a - b < x < b - a \\ a + x & ; x \ge b - a \end{cases}$$

By Quick Concept

$$f'(x) = \begin{cases} -1 & ; x \le a - b \\ 0 & ; a - b < x < b - a \\ 1 & ; x \ge b - a \end{cases}$$

hence not differentiable at two points.

Q59. Column-I contains 4 functions and column-II contains comments w.r.t their continuity and differentiability at x = 0. Note that column-I may have more than one matching options in column-II.

Column-I

Column-II

(A) $f(x) = [x] + |1 - x|$ [ ] denotes the greatest integer function

(P) continuous

(B) $g(x) = |x - 2| + |x|$

(Q) differentiable

(C) $h(x) = [\tan^2 x]$ [ ] denotes the greatest integer function

(R) discontinuous

(D) $l(x) = \begin{cases} \dfrac{x(3e^{1/x}+4)}{(2-e^{e/x})} & x \neq 0 \\ 0 & x = 0 \end{cases}$ (S) non derivable

(A) (A) R, S; (B) P, S; (C) P, Q; (D) P, S (B) (A) R, P; (B) P, S; (C) P, Q; (D) P, Q

(C) (A) R, S; (B) P, Q; (C) P, R; (D) P, S (D) (A) Q, S; (B) P, R; (C) R, Q; (D) P, S

Ans. 5f166817b7c373b2a599afeb

Sol. (A) $f(x) = [x] + |1 - x|$ at $x = 0$

Continuity

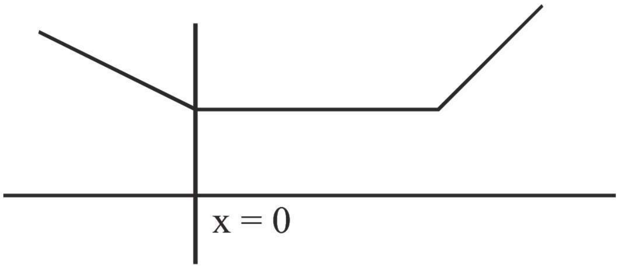
R.H.L. $= [0 + h] + |1 - 0 - h| = 0 + |1 - h| = 1$

L.H.L. $= [0 - h] + |1 - 0 + h| = -1 + 1 = 0$

hence not continuous and not derivable.

(B) $= g(x) = |x - 2| + |x|$

Graph



Clearly it is continuous at x = 0 but not derivable.

(C) $h(x) = [\tan^2 x]$

$h(0) = 0$

R.H.L. = 0

L.H.L. = 0

differentiability

$$\mathrm{R.H.D.} = \lim_{h\to 0}\frac{f(0+h)-f(0)}{h}$$

$$= \lim_{h\to 0}\frac{0-0}{h} = 0$$

$$\mathrm{L.H.D.} = \lim_{h\to 0}\frac{f(0-h)-f(0)}{-h} = 0$$

hence continuous and derivable.

(D) at x = 0

$$\mathrm{R.H.L.} = \frac{h\left(3^{\frac{1}{h}}+4\right)}{\left(2-e^{\frac{1}{h}}\right)} = \frac{h\left(3+4e^{\frac{1}{h}}\right)}{\left(2e^{\frac{1}{h}}-1\right)} = 0$$

$$\mathrm{L.H.L.} = \frac{(-h)\left(3e^{\frac{1}{h}}+h\right)}{\left(2-e^{\frac{1}{h}}\right)} = 0$$

$f(0) = 0$

hence continuous at x = 0.

differentiability

$$\mathrm{R.H.D.} = \lim_{h\to 0}\frac{f(0+h)-f(0)}{h} = \frac{h\left(3e^{\frac{1}{h}}+4\right)}{\frac{2-e^{\frac{1}{h}}}{h}} = \frac{\left(3+4e^{\frac{1}{h}}\right)}{\left(2e^{\frac{1}{h}}-1\right)} = -3$$

$$\mathrm{L.H.L.} = \frac{(-h)\left(3e^{\frac{1}{h}}+h\right)}{\frac{\left(2-e^{\frac{e}{h}}\right)}{-h}} = 2$$

correct option = (P) & (S)

Q60. Column-I

Column-II

(A) $f(x) = \begin{bmatrix} x+1 & if & x<0 \\ \cos & if & x \geq 0 \end{bmatrix}$ at x = 0 is (P)

continuous

(B) For every x ∈ R the function g (x) = $\frac{\sin(\pi[x-\pi])}{1+[x]^2}$ (Q) derivable

where [x] denotes the greatest integer function is (R) discontinuous

(C) h (x) = $\sqrt{\{x\}^2}$where {x} denotes fractional part function ∀ x ∈ I, is (S) non derivable

(D) k (x) = $\begin{bmatrix} x^{\frac{1}{lnx}} & if & x \neq 1 \\ e & if & x = 1 \end{bmatrix}$ at x = 1 is

(A) (A) P, Q; (B) P, S; (C) R, S; (D) P, R

(B) (A) P, S; (B) P, Q; (C) R, S; (D) P, Q

(C) (A) Q, S; (B) P, R; (C) P, S; (D) P, Q

(D) (A) P, Q; (B) P, S; (C) R, Q; (D) Q, S

Ans. 5f166cf06c0af9b25a96aa3e

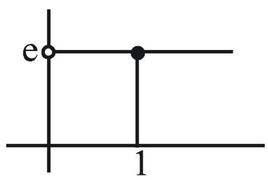
Sol. (A) $f'(0) = \lim_{h\to 0} \frac{cos\,h - 0}{h}$ does not exist.

Obviously f (0) = f (0^-) = f (0^+) = 1

Hence continuous and not derivable

(B) g (x) = 0 for all x, hence continuous and derivable

(C) as 0 ≤ { f (x) } < 1, hence h (x) = $\sqrt{(x)^2}$ = {x} which is discontinuous hence non derivable all x ∈ I



(D) $\lim_{x\to 1} x^{\frac{1}{lnx}} = \lim_{x\to 1} x^{log_x e} = e = f(1)$

Hence k (x) is constant for all x > 0 hence continuous and differentiable at x = 1.

Q61. Let f(x) = $\begin{cases} [x], & -2 \le x < 0 \\ |x| & 0 \le x \le 2 \end{cases}$ (where [.] denotes the greatest ineger function) g(x) = sec x, x ∈ R – (2n + 1) $\frac{\pi}{2}$, n ∈ I

Match the following statements in column I with their values in column II in the interval $\left(-\frac{3\pi}{2}, \frac{3\pi}{2}\right)$.

Column - I

Column-II

(A) Abscissa of points where limit of fog(x) exist is/are

(P) –1

(B) Abscissa points in domain of gof(x), where limit of gof(x) does not

(Q) π

exist is/are

(C) Abscissa of point of discontinuity of fog (x) is/are

(R) $\frac{5\pi}{6}$

(D) Abscissa of points of differentiability of fog(x) is/are

(S) $-\pi$

(T) 0

(A) (A) RS, (B) PT, (C) PQRS, (D) PRT

(B) (A) PQRST, (B) PT, (C) PS, (D) PQ

(C) (A) QT, (B) PQRST, (C) QS, (D) PRT

(D) (A) PQRST, (B) PT, (C) QS, (D) PRT

Ans. 5f1670aa6c0af9b25a96ab77

Sol. -